

Assignment 7: GLMs (Linear Regressios, ANOVA, & t-tests)

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Fall 2025

OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on generalized linear models.

Directions

1. Rename this file `<FirstLast>_A07_GLMs.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure to **answer the questions** in this assignment document.
5. When you have completed the assignment, **Knit** the text and code into a single PDF file.

!! AI generated code should NOT be used in this assignment

Set up your session

1. Set up your session. Check your working directory. Load the tidyverse, agricolae and other needed packages. Import the *raw* NTL-LTER raw data file for chemistry/physics (NTL-LTER_Lake_ChemistryPhysics_Raw.csv). Set date columns to date objects.
2. Build a ggplot theme and set it as your default theme.

```
#1  
getwd()
```

```
## [1] "/Users/mayike/Desktop/Duke/EDE_Fall2025"
```

```
library(tidyverse)
```

```
## Warning: package 'ggplot2' was built under R version 4.4.1
```

```
## Warning: package 'tidyr' was built under R version 4.4.1
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --  
## v dplyr      1.1.4      v readr      2.1.5  
## v forcats    1.0.0      v stringr    1.5.1  
## v ggplot2    3.5.2      v tibble     3.2.1
```

```
## v lubridate 1.9.3      v tidyr      1.3.1
## v purrr      1.0.2
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(agricolae)
library(lubridate)
raw <- read.csv("./Data/Raw/NTL-LTER_Lake_ChemistryPhysics_Raw.csv", stringsAsFactors = T)
raw$sampleddate <- mdy(raw$sampleddate)
#2
mytheme <- theme_classic(base_size = 14) +
  theme(axis.text = element_text(color = "black"),
        legend.position = "top")
theme_set(mytheme)
```

Simple regression

Our first research question is: Does mean lake temperature recorded during July change with depth across all lakes?

3. State the null and alternative hypotheses for this question: > Answer: H0: There are no relationship between lake temperature and depth, and mean lake temperature is equal across all lakes. Ha: There is a linear relationship such that lake temperature decreases with lake depth.
4. Wrangle your NTL-LTER dataset with a pipe function so that the records meet the following criteria:
 - Only dates in July.
 - Only the columns: lakename, year4, daynum, depth, temperature_C
 - Only complete cases (i.e., remove NAs)
5. Visualize the relationship among the two continuous variables with a scatter plot of temperature by depth. Add a smoothed line showing the linear model, and limit temperature values from 0 to 35 °C. Make this plot look pretty and easy to read.

```
#4
ntl <- raw %>%
  filter(month(sampleddate) == 7) %>%
  select(c(lakename:temperature_C), -sampledate) %>%
  na.omit()

summary(ntl)
```

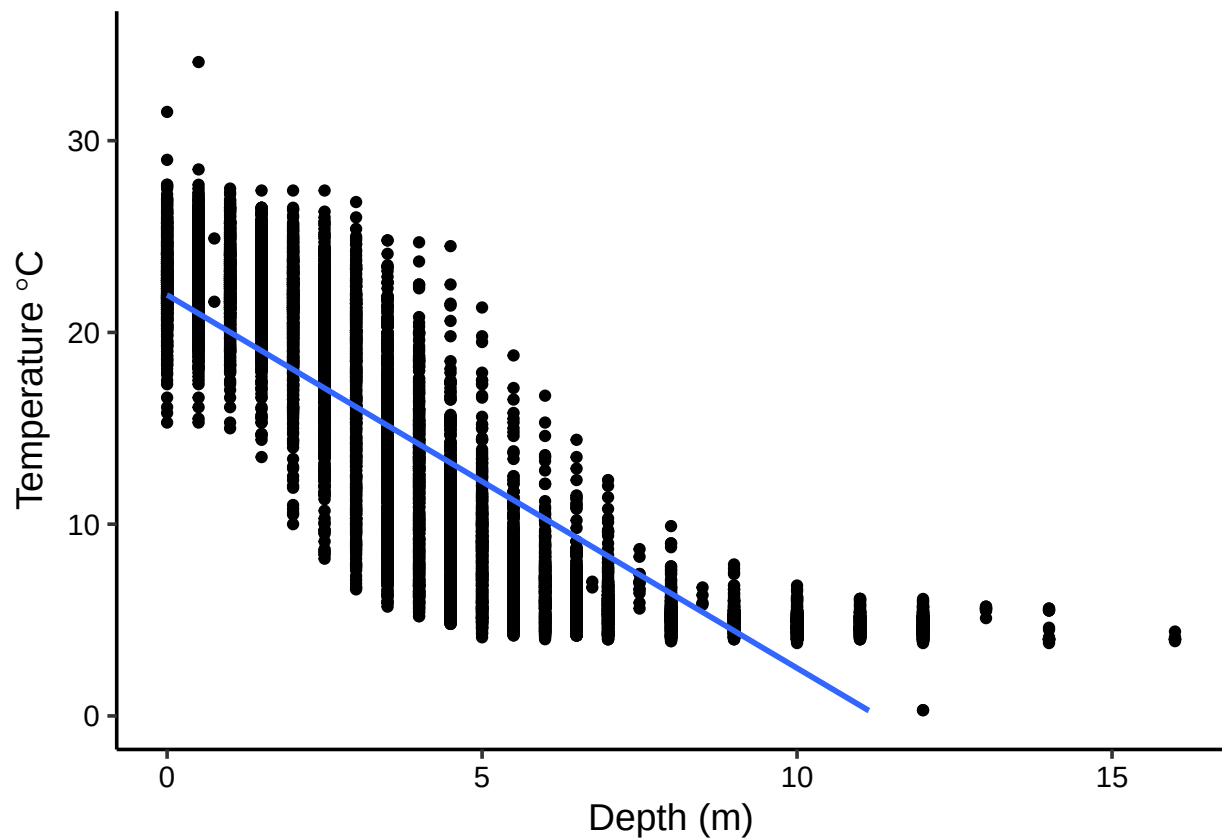
##		lakename	year4	daynum	depth
##	Peter Lake	:2872	Min. :1984	Min. :182.0	Min. : 0.000
##	Paul Lake	:2660	1st Qu.:1992	1st Qu.:190.0	1st Qu.: 2.000
##	Tuesday Lake	:1524	Median :1998	Median :198.0	Median : 4.500
##	West Long Lake	:1026	Mean :1999	Mean :197.5	Mean : 4.745
##	East Long Lake	: 968	3rd Qu.:2006	3rd Qu.:205.0	3rd Qu.: 7.000
##	Crampton Lake	: 318	Max. :2016	Max. :213.0	Max. :16.000

```
## (Other)      : 360
## temperature_C
## Min.       : 0.30
## 1st Qu.: 5.50
## Median :10.10
## Mean      :12.72
## 3rd Qu.:20.80
## Max.       :34.10
##
```

```
#5
temp_dep <- ntl %>%
  ggplot(aes(x = depth, y = temperature_C))+
  geom_point()+
  geom_smooth(method = "lm")+
  ylim(c(0,35))+
  xlab("Depth (m)") +
  ylab(expression("Temperature" ~ degree "C"))
temp_dep
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 24 rows containing missing values or values outside the scale range
## ('geom_smooth()').
```



6. Interpret the figure. What does it suggest with regards to the response of temperature to depth? Do the distribution of points suggest about anything about the linearity of this trend?

Answer: The temperature of the lake decreases with depth. The linear regression model suggests there is a negative linear trend, however, there is a deviation of data points when the depth is above 10m.

7. Perform a linear regression to test the relationship and display the results.

```
#7
lm <- lm(data = ntl, temperature_C~depth)
summary(lm)

##
## Call:
## lm(formula = temperature_C ~ depth, data = ntl)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -9.5173 -3.0192  0.0633  2.9365 13.5834
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  21.95597   0.06792   323.3  <2e-16 ***
## depth       -1.94621   0.01174  -165.8  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.835 on 9726 degrees of freedom
## Multiple R-squared:  0.7387, Adjusted R-squared:  0.7387
## F-statistic: 2.75e+04 on 1 and 9726 DF, p-value: < 2.2e-16
```

8. Interpret your model results in words. Include how much of the variability in temperature is explained by changes in depth, the degrees of freedom on which this finding is based, and the statistical significance of the result. Also mention how much temperature is predicted to change for every 1m change in depth.

Answer: The variation of the linear model (R-squared) is 0.7387. F value is 2.75e+04 and p-value < 2.2e-16, indicating this model is statistically significant. The intercept of the linear regression model appears to be 21.96 and a slope of -1.95, indicating every 1m deeper in depth results in 1.95 degree Celcius of decrease.

Multiple regression

Let's tackle a similar question from a different approach. Here, we want to explore what might the best set of predictors for lake temperature in July across the monitoring period at the North Temperate Lakes LTER.

9. Run an AIC to determine what set of explanatory variables (year4, daynum, depth) is best suited to predict temperature.

10. Run a multiple regression on the recommended set of variables.

```
#9
mlr <- lm(data = ntl, temperature_C~year4+daynum+depth)
step(mlr)

## Start:  AIC=26065.53
## temperature_C ~ year4 + daynum + depth
##
##           Df Sum of Sq    RSS   AIC
## <none>                 141687 26066
## - year4    1         101 141788 26070
## - daynum   1         1237 142924 26148
## - depth    1      404475 546161 39189

##
## Call:
## lm(formula = temperature_C ~ year4 + daynum + depth, data = ntl)
##
## Coefficients:
## (Intercept)      year4      daynum      depth
##   -8.57556    0.01134    0.03978   -1.94644
```

```
#10
mlr <- lm(data = ntl, temperature_C~year4+daynum+depth)
summary(mlr)

##
## Call:
## lm(formula = temperature_C ~ year4 + daynum + depth, data = ntl)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -9.6536 -3.0000  0.0902  2.9658 13.6123
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -8.575564   8.630715  -0.994  0.32044
## year4        0.011345   0.004299   2.639  0.00833 **
## daynum       0.039780   0.004317   9.215 < 2e-16 ***
## depth       -1.946437   0.011683 -166.611 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.817 on 9724 degrees of freedom
## Multiple R-squared:  0.7412, Adjusted R-squared:  0.7411
## F-statistic: 9283 on 3 and 9724 DF, p-value: < 2.2e-16
```

11. What is the final set of explanatory variables that the AIC method suggests we use to predict temperature in our multiple regression? How much of the observed variance does this model explain? Is this an improvement over the model using only depth as the explanatory variable?

Answer: AIC suggests to use year4, daynum and depth as variables to predict the multiple linear regression model. The variance is 0.7412 here, which demonstrates an improvement than single linear regression model(0.7387).

Analysis of Variance

12. Now we want to see whether the different lakes have, on average, different temperatures in the month of July. Run an ANOVA test to complete this analysis. (No need to test assumptions of normality or similar variances.) Create two sets of models: one expressed as an ANOVA model and another expressed as a linear model (as done in our lessons).

#12

```
temp_anova <- aov(data = ntl, temperature_C ~ lakename)
temp_anova
```

```
## Call:
## aov(formula = temperature_C ~ lakename, data = ntl)
##
## Terms:
##          lakename Residuals
## Sum of Squares  21641.9  525813.4
## Deg. of Freedom      8      9719
##
## Residual standard error: 7.355379
## Estimated effects may be unbalanced
```

```
summary(temp_anova)
```

```
##          Df Sum Sq Mean Sq F value Pr(>F)
## lakename    8  21642   2705.2     50 <2e-16 ***
## Residuals  9719 525813    54.1
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
temp_lm <- lm(data = ntl, temperature_C ~ lakename)
temp_lm
```

```
##
## Call:
## lm(formula = temperature_C ~ lakename, data = ntl)
##
## Coefficients:
##          (Intercept)      lakenameCrampton Lake      lakenameEast Long Lake
##              17.666                -2.315                -7.399
## lakenameHummingbird Lake      lakenamePaul Lake      lakenamePeter Lake
##             -6.893                -3.852                -4.350
##      lakenameTuesday Lake      lakenameWard Lake      lakenameWest Long Lake
##             -6.597                -3.208                -6.088
```

```
summary(temp_lm)
```

```
##
## Call:
## lm(formula = temperature_C ~ lakename, data = ntl)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10.769   -6.614   -2.679    7.684   23.832
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      17.6664     0.6501  27.174 < 2e-16 ***
## lakenameCrampton Lake    -2.3145     0.7699   -3.006 0.002653 **
## lakenameEast Long Lake   -7.3987     0.6918 -10.695 < 2e-16 ***
## lakenameHummingbird Lake -6.8931     0.9429   -7.311 2.87e-13 ***
## lakenamePaul Lake        -3.8522     0.6656   -5.788 7.36e-09 ***
## lakenamePeter Lake       -4.3501     0.6645   -6.547 6.17e-11 ***
## lakenameTuesday Lake    -6.5972     0.6769   -9.746 < 2e-16 ***
## lakenameWard Lake        -3.2078     0.9429   -3.402 0.000672 ***
## lakenameWest Long Lake   -6.0878     0.6895   -8.829 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.355 on 9719 degrees of freedom
## Multiple R-squared:  0.03953,    Adjusted R-squared:  0.03874
## F-statistic:    50 on 8 and 9719 DF,  p-value: < 2.2e-16
```

13. Is there a significant difference in mean temperature among the lakes? Report your findings.

Answer: We obtained a p value of $<2e-16$ here, which rejects the null hypothesis, meaning mean temperature are significantly different among the lakes. But we cannot tell which lake is significantly different to another merely running anova test.

14. Create a graph that depicts temperature by depth, with a separate color for each lake. Add a `geom_smooth` (method = "lm", se = FALSE) for each lake. Make your points 50 % transparent. Adjust your y axis limits to go from 0 to 35 degrees. Clean up your graph to make it pretty.

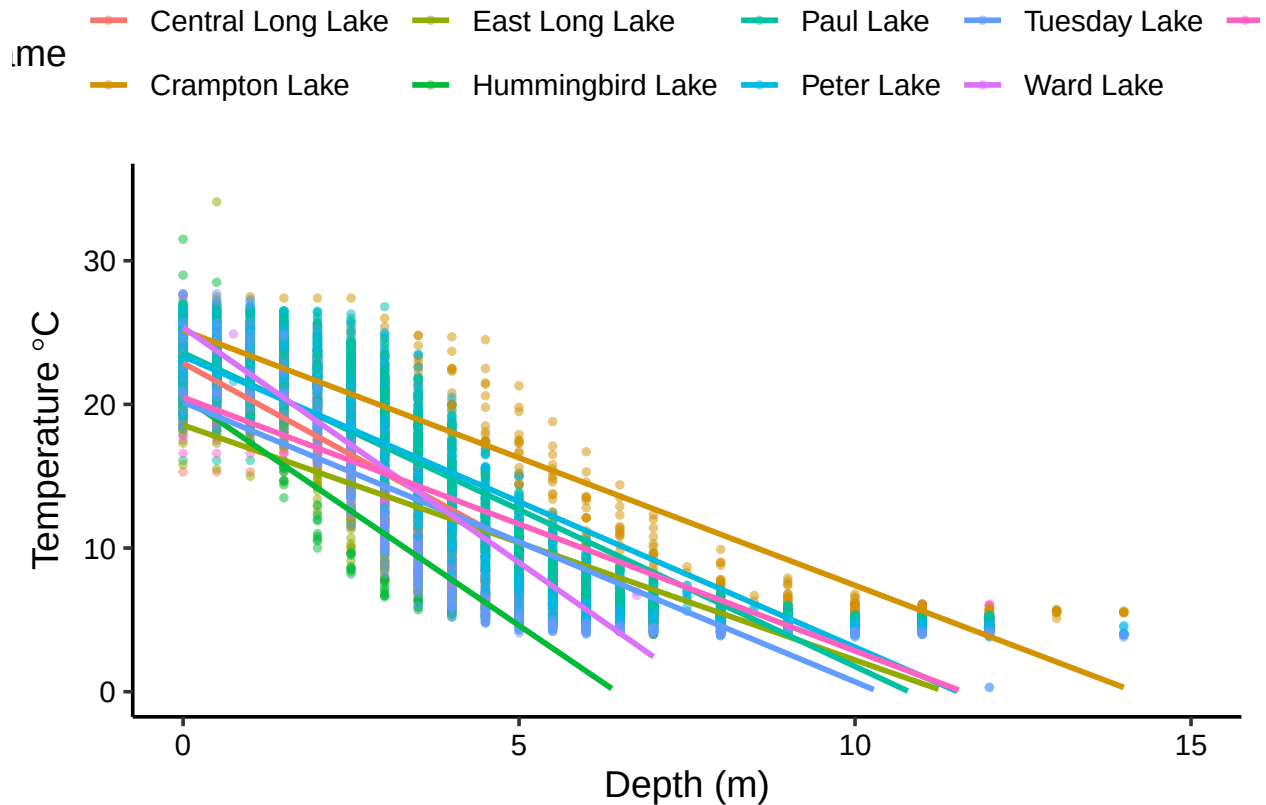
```
#14.
ntl_lake <- ntl %>%
  ggplot(aes(x = depth, y = temperature_C, colour = lakename))+
  geom_point(alpha = 0.5, size = 1)+
  geom_smooth(method = "lm", se = F)+
  ylim(c(0,35))+
  xlim(c(0,15))+
  xlab("Depth (m)") +
  ylab(expression("Temperature" ~ degree ~ "C")) +
  labs(color = "Lake Name")
ntl_lake
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 9 rows containing non-finite outside the scale range
## ('stat_smooth()').

## Warning: Removed 9 rows containing missing values or values outside the scale range
## ('geom_point()').

## Warning: Removed 58 rows containing missing values or values outside the scale range
## ('geom_smooth()').
```



15. Use the Tukey's HSD test to determine which lakes have different means.

#15

```
TukeyHSD(temp_anova)
```

```
## Tukey multiple comparisons of means
## 95% family-wise confidence level
##
## Fit: aov(formula = temperature_C ~ lakename, data = ntl)
##
## $lakename
##
```

	diff	lwr	upr	p adj
Crampton Lake-Central Long Lake	-2.3145195	-4.7031913	0.0741524	0.0661566
East Long Lake-Central Long Lake	-7.3987410	-9.5449411	-5.2525408	0.0000000
Hummingbird Lake-Central Long Lake	-6.8931304	-9.8184178	-3.9678430	0.0000000


```
## Paul Lake-Central Long Lake      -3.8521506 -5.9170942 -1.7872070 0.0000003
## Peter Lake-Central Long Lake      -4.3501458 -6.4115874 -2.2887042 0.0000000
## Tuesday Lake-Central Long Lake    -6.5971805 -8.6971605 -4.4972005 0.0000000
## Ward Lake-Central Long Lake       -3.2077856 -6.1330730 -0.2824982 0.0193405
## West Long Lake-Central Long Lake  -6.0877513 -8.2268550 -3.9486475 0.0000000
## East Long Lake-Crampton Lake      -5.0842215 -6.5591700 -3.6092730 0.0000000
## Hummingbird Lake-Crampton Lake    -4.5786109 -7.0538088 -2.1034131 0.0000004
## Paul Lake-Crampton Lake           -1.5376312 -2.8916215 -0.1836408 0.0127491
## Peter Lake-Crampton Lake          -2.0356263 -3.3842699 -0.6869828 0.0000999
## Tuesday Lake-Crampton Lake        -4.2826611 -5.6895065 -2.8758157 0.0000000
## Ward Lake-Crampton Lake           -0.8932661 -3.3684639  1.5819317 0.9714459
## West Long Lake-Crampton Lake      -3.7732318 -5.2378351 -2.3086285 0.0000000
## Hummingbird Lake-East Long Lake   0.5056106 -1.7364925  2.7477137 0.9988050
## Paul Lake-East Long Lake          3.5465903  2.6900206  4.4031601 0.0000000
## Peter Lake-East Long Lake         3.0485952  2.2005025  3.8966879 0.0000000
## Tuesday Lake-East Long Lake       0.8015604 -0.1363286  1.7394495 0.1657485
## Ward Lake-East Long Lake          4.1909554  1.9488523  6.4330585 0.0000002
## West Long Lake-East Long Lake     1.3109897  0.2885003  2.3334791 0.0022805
## Paul Lake-Hummingbird Lake        3.0409798  0.8765299  5.2054296 0.0004495
## Peter Lake-Hummingbird Lake       2.5429846  0.3818755  4.7040937 0.0080666
## Tuesday Lake-Hummingbird Lake     0.2959499 -1.9019508  2.4938505 0.9999752
## Ward Lake-Hummingbird Lake        3.6853448  0.6889874  6.6817022 0.0043297
## West Long Lake-Hummingbird Lake   0.8053791 -1.4299320  3.0406903 0.9717297
## Peter Lake-Paul Lake              -0.4979952 -1.1120620  0.1160717 0.2241586
## Tuesday Lake-Paul Lake            -2.7450299 -3.4781416 -2.0119182 0.0000000
## Ward Lake-Paul Lake               0.6443651 -1.5200848  2.8088149 0.9916978
## West Long Lake-Paul Lake          -2.2356007 -3.0742314 -1.3969699 0.0000000
## Tuesday Lake-Peter Lake           -2.2470347 -2.9702236 -1.5238458 0.0000000
## Ward Lake-Peter Lake              1.1423602 -1.0187489  3.3034693 0.7827037
## West Long Lake-Peter Lake         -1.7376055 -2.5675759 -0.9076350 0.0000000
## Ward Lake-Tuesday Lake            3.3893950  1.1914943  5.5872956 0.0000609
## West Long Lake-Tuesday Lake       0.5094292 -0.4121051  1.4309636 0.7374387
## West Long Lake-Ward Lake          -2.8799657 -5.1152769 -0.6446546 0.0021080
```

```
ntl.groups <- HSD.test(temp_anova, "lakename", group = TRUE)
ntl.groups
```

```
## $statistics
##      MSerror Df      Mean      CV
##    54.1016 9719 12.72087 57.82135
##
## $parameters
##      test  name.t ntr StudentizedRange alpha
##    Tukey lakename  9      4.387504  0.05
##
## $means
##               temperature_C      std      r      se Min  Max   Q25   Q50
## Central Long Lake      17.66641 4.196292  128 0.6501298 8.9 26.8 14.400 18.40
## Crampton Lake          15.35189 7.244773  318 0.4124692 5.0 27.5  7.525 16.90
## East Long Lake         10.26767 6.766804  968 0.2364108 4.2 34.1  4.975  6.50
## Hummingbird Lake       10.77328 7.017845  116 0.6829298 4.0 31.5  5.200  7.00
## Paul Lake              13.81426 7.296928 2660 0.1426147 4.7 27.7  6.500 12.40
## Peter Lake              13.31626 7.669758 2872 0.1372501 4.0 27.0  5.600 11.40
## Tuesday Lake           11.06923 7.698687 1524 0.1884137 0.3 27.7  4.400  6.80
```

```
## Ward Lake          14.45862 7.409079  116 0.6829298 5.7 27.6  7.200 12.55
## West Long Lake     11.57865 6.980789 1026 0.2296314 4.0 25.7  5.400  8.00
##                      Q75
## Central Long Lake  21.000
## Crampton Lake      22.300
## East Long Lake     15.925
## Hummingbird Lake   15.625
## Paul Lake          21.400
## Peter Lake         21.500
## Tuesday Lake       19.400
## Ward Lake          23.200
## West Long Lake     18.800
##
## $comparison
## NULL
##
## $groups
##           temperature_C groups
## Central Long Lake      17.66641      a
## Crampton Lake          15.35189     ab
## Ward Lake              14.45862     bc
## Paul Lake              13.81426      c
## Peter Lake             13.31626      c
## West Long Lake         11.57865      d
## Tuesday Lake           11.06923     de
## Hummingbird Lake       10.77328     de
## East Long Lake         10.26767      e
##
## attr(,"class")
## [1] "group"
```

16. From the findings above, which lakes have the same mean temperature, statistically speaking, as Peter Lake? Does any lake have a mean temperature that is statistically distinct from all the other lakes?

Answer: Paul Lake and Ward Lake statistically share same mean temperature as Peter lake. Mean temperature of Central Long Lake and Crampton Lake are significantly different from others. However, there is no lake statistically distinct from all the other lakes in this example.

17. If we were just looking at Peter Lake and Paul Lake. What's another test we might explore to see whether they have distinct mean temperatures?

Answer: We can run two sample t-test.

18. Wrangle the July data to include only records for Crampton Lake and Ward Lake. Run the two-sample T-test on these data to determine whether their July temperature are same or different. What does the test say? Are the mean temperatures for the lakes equal? Does that match you answer for part 16?

```
ntl.Crampton.Ward <- ntl %>%
  filter(lakename %in% c("Crampton Lake" , "Ward Lake"))

t.test(ntl.Crampton.Ward$temperature_C~ntl.Crampton.Ward$lakename)
```

```
##
## Welch Two Sample t-test
##
## data: ntl.Crampton.Ward$temperature_C by ntl.Crampton.Ward$lakename
## t = 1.1181, df = 200.37, p-value = 0.2649
## alternative hypothesis: true difference in means between group Crampton Lake and group Ward Lake is not equal to 0
## 95 percent confidence interval:
## -0.6821129 2.4686451
## sample estimates:
## mean in group Crampton Lake      mean in group Ward Lake
##                15.35189                14.45862
```

Answer: T-test suggests a p-value of 0.26 (<0.5), so the mean temperature in both lakes are different. Mean temperature for Crampton Lake and Ward Lake is 15.35 and 14.46 degree Celcius, respectively. However, the mean values were shown statistically the same in question 16, which does not match, this could be caused by the larger categorical variables in question 16.